

Ung Heo

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Bionics Research Center,
Korea Institute of Science and Technology
(KIST)

EDUCATION

- 2025 Ph.D. in Mechanical Engineering
Korea Advanced Institute of Science and Technology
Dissertation: **Chemical Reaction-based Mobile Pneumatic Source for Wearable Robots**
Advisor: Prof. Jung Kim
Keyword: Soft wearable robots, Pneumatic actuation, Self-temperature regulation
- 2020 M.S. in Mechanical Engineering
Korea Advanced Institute of Science and Technology
Thesis: **Portable Pneumatic Back Support Exoskeleton and Assistive Strategy for Lifting Assistance**
Advisor: Prof. Jung Kim
Keyword: Industrial exoskeleton, Low back pain, Surface electromyography
- 2018 B.S. in Mechanical Engineering
Korea Advanced Institute of Science and Technology
GPA: 3.83/4.3 (Cum Laude)

PROFESSIONAL EXPERIENCE

- July 2025 – Current Postdoctoral Researcher, Bionics Research Center,
Korea Institute of Science and Technology (KIST)
- Mar 2025 – Jun 2025 Postdoctoral Researcher, Mechanical Engineering Research Institute
Korea Advanced Institute of Science and Technology (KAIST)

PROJECTS

- Mar 2026 – Current **Development of a chemical-based back support exoskeleton**
- Project objective: Chemically-driven silent back support exoskeleton
 - Funding: Ministry of Science and ICT (MSIT)
 - Role & Contribution: Developing a chemical-based pneumatic power source, back support exoskeleton, and experimental platform
- Jun 2022 – Jun 2025 **Development of a wearable sensor for exosuit control**
- Collaboration: Korea Institute of Machinery & Materials (KIMM)
 - Project objective: Biosignal-based control of an exosuit for elderly individuals
 - Funding: Ministry of Science and ICT (MSIT)
 - Role & Contribution: Developed an integrated bioimpedance-sEMG sensor, performed biosignal data acquisition, and conducted signal processing for a deep learning-based gait classification system

- Dec 2022 – Jun 2025 **Development of a pneumatic actuation system for drone perching**
- Project objective: Design and development of a drone perching mechanism using soft pneumatic grippers for extended military operations
 - Funding: Defense Acquisition Program Administration (DAPA)
 - Role & Contribution: Designed the pneumatic actuation system for soft grippers
- Aug 2021 – Dec 2022 **Ergonomic chair design with musculoskeletal simulation**
- Project objective: Design of a comfortable chair frame and tilting mechanism
 - Funding: Ceragem
 - Role & Contribution: Designed the chair tilting mechanism, conducted musculoskeletal simulations using AnyBody software, and performed physiological measurements for evaluation
- Sept 2020 – Dec 2022 **Robotic prosthesis control with biosignals**
- Collaboration: Seoul National University Hospital (Prof. Dong Yeon Lee)
 - Project objective: Control of a robotic prosthesis using an integrated bioimpedance-sEMG biosignal sensor
 - Funding: Ministry of Science and ICT (MSIT)
 - Role & Contribution: **Project manager**; developed the biosignal sensor, deep learning-based real-time gait classifier, and robotic prosthesis controller
- Mar 2018 – Jun 2022 **Mobile pneumatic ankle-foot orthosis for drop foot patients**
- Collaboration: Seoul National University Hospital (Prof. Dong Yeon Lee), Korea Institute of Industrial Technology (KITECH)
 - Project objective: Development of a pneumatic ankle-foot orthosis (AFO) for patients with drop foot
 - Funding: Ministry of Science and ICT (MSIT)
 - Role & Contribution: Enhanced the design of the pneumatic pump and AFO; conducted patient experiments for performance evaluation
- May 2019 – Dec 2019 **Pneumatic back support exoskeleton for laborers**
- Project objective: Development of a pneumatic back support exoskeleton and assistive strategies to prevent low back pain
 - Funding: Korea Advanced Institute of Science and Technology
 - Role & Contribution: **Project manager**; developed the back support exoskeleton, implemented sEMG-based lifting detection, and conducted human experiments for evaluation
- Jun 2018 – Dec 2018 **sEMG-controlled lower-limb exoskeleton**

- Project objective: Robust intention detection for lower-limb exoskeleton control using surface electromyography (sEMG)
- Funding: Defense Acquisition Program Administration (DAPA)
- Role & Contribution: Designed the lower-limb exoskeleton; developed and implemented sEMG signal processing for control and performance evaluation

PUBLICATIONS

Journal Publications

- [1] **Ung Heo**, Junhwan Choi, Sangjoon J. Kim, and Jung Kim. "Low-heat and Near-Silent Pneumatic Source Driven by Integrated Endothermic–Exothermic Chemical Reactions for Soft Robots." *Soft Robotics*, 2026
- [2] Heesoo Kim, Juhyun Lee, **Ung Heo**, Dhileep Kumar Jayashankar, Karen-Christian Agno, Yeji Kim, Choong Yeon Kim et al. "Skin preparation–free, stretchable microneedle adhesive patches for reliable electrophysiological sensing and exoskeleton robot control. *Science Advances*, 2024
- [3] **Ung Heo**, Jirou Feng, Sangjoon J. Kim, and Jung Kim. "sEMG-triggered fast assistance strategy for a pneumatic back support exoskeleton." *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2022
- [4] **Ung Heo**, Sangjoon J. Kim, and Jung Kim. "Backdrivable and fully-portable pneumatic back support exoskeleton for lifting assistance." *IEEE Robotics and Automation Letters*, 2020
- [5] Seulki Kyeong, Wonseok Shin, Minjin Yang, **Ung Heo**, Jirou Feng, and Jung Kim. "Recognition of walking environments and gait period by surface electromyography." *Frontiers of Information Technology & Electronic Engineering*, 2019

Journal Papers in Review or in Preparation

- [1] **Ung Heo**, Junhwan Choi, Seongjong Shin, Youngsuk Nam, Sangjoon J. Kim, and Jung Kim. "Electronics-free, self-temperature-regulating pneumatic source for robotic applications." *IEEE/ASME Transactions on Mechatronics* (Under review).
- [2] **Ung Heo**, Sangjoon J. Kim. "A closed-loop phase-change pneumatic source enabling bidirectional pressure generation for soft actuation" (In preparation).
- [3] **Ung Heo**, Sangjoon J. Kim. "A back support exoskeleton based on a closed-loop phase-change pneumatic source" (In preparation).

Conference Papers

[1] **Ung Heo**, Hyunho Jeong, Jirou Feng, Junhwi Cho, Kyungseo Park, Youngsik Yoon, Dongyeon Lee, and Jung Kim. "Development of a Bioimpedance and sEMG Fusion Sensor for Gait Phase Detection: Validation with a Transtibial Amputee." In *2023 45th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, 2023. (Oral presentation)

[2] **Ung Heo**, Sangjoon J. Kim, and Jung Kim. "Backdrivable and fully-portable pneumatic back support exoskeleton for lifting assistance." In *2020 IEEE International Conference on Robotics and Automation (ICRA)*, (with journal option) 2020

Conference Abstracts and Presentations

[1] **Ung Heo**, Jung Kim, and Sangjoon J. Kim. "Self-regulating low-noise mobile pneumatic source for wearable robots in diverse environments." IFAC, 2026. (accepted)

[2] **Ung Heo**, Jung Kim, and Sangjoon J. Kim. "Evaluating portable pneumatic power sources for wearable robots: flow density and acoustic noise comparison." ICNR, 2026. (accepted)

[3] **Ung Heo**, Sangjoon J. Kim, and Jung Kim. "Experimental Evaluation of Various Pneumatic Generators for Pneumatically Driven Wearable Robots." Korean Society of Biomechanics, 2025.

[4] **Ung Heo**, and Jung Kim. "Hybrid Wearable Sensor using Hall Sensor and sEMG for Joint Dynamics Estimation." IEEE-EMBS BSN, 2024.

[5] **Ung Heo**, and Jung Kim. "Development and Control Method of Chemical Reaction-based Pneumatic Source for Wearable Robots." ICROS, 2024.

[6] Woojong Kim, **Ung Heo**, Sangha Jeon, and Jung Kim. "Development of a Fabric-based Pneumatic Bistable Gripper for Drone Perching." ICROS, 2024.

[7] **Ung Heo**, and Jung Kim. "sEMG-Hall Sensor Integrated Wearable Sensor for Joint Dynamics Estimation." Korean Society of Biomechanics, 2024.

[8] **Ung Heo**, Hyunho Jeong, Chaeree Park, Dayoung Yeom, and Jung Kim. "Gait Phase Estimation for Robotic Prosthesis Control based on Bioimpedance." KRoC, 2023.

[9] **Ung Heo**, and Jung Kim. "Wearable Bioimpedance and Electromyography Integrated Sensor for Motion Intention Detection." Korean Society of Biomechanics, 2021.

[10] **Ung Heo**, and Jung Kim. "Evaluation of sEMG-Triggered Rapid Assistance in Back Support Exoskeleton." KSPE Conference, 2021.

PATENTS

[1] Jung Kim, **Ung Heo**. "Pneumatic pressure generator and pneumatic actuator based on endothermic chemical reaction", Korean Patent, 10-2936054 (2026), granted.

[2] Jung Kim, **Ung Heo**, and Junhwan Choi. "Portable pneumatic generator and wearable robot including same", US Patent, 19199248 (2025), patent application has been done.

[3] Jung Kim, **Ung Heo**, and Junhwan Choi. "Self-temperature-regulated mobile pneumatic source", Korean Patent, 10-2025-0131901 (2025), patent application has been done.

[4] Jung Kim, **Ung Heo**, and Junhwan Choi. "Portable pneumatic generator and wearable robot including same", Korean Patent, 10-2024-0110980 (2024), patent application has been done.

AWARDS

[1] 2025 Outstanding Paper Award for Oral Presentation, Korean Society of Biomechanics

LANGUAGES

Korean: *Native Speaker*

English: *Fluent*

TEACHING AND MENTORING

Fall semester, 2023	Teaching Assistant for Human and Machine •Lecture assistance
Spring semester, 2023	Teaching Assistant for Applied Electronics •Experimental session on Wheatstone bridge and load cell
Fall semester, 2022	Teaching Assistant for Human and Machine •Lecture assistance
Spring semester, 2022	Teaching Assistant for Basic Mechanical Practice •Experimental session on machining
Fall semester, 2021	Teaching Assistant for Mechatronic System Design •Experimental session on AI-based autonomous driving
Spring semester, 2021	Teaching Assistant for Basic Mechanical Practice / Counseling Assistant •3D/2D CAD drawing

- Counseling for graduate and undergraduate students
- Fall semester, 2020 Teaching Assistant for Modeling and Control of Engineering Systems
- Lecture assistance
- Spring semester, 2020 Teaching Assistant for Automatic Control / Counseling Assistant
- Lecture assistance
 - Counseling for graduate and undergraduate students

TECHNICAL SKILLS

Programming: MATLAB, Python, C (Arduino)

Data acquisition and control: LabVIEW FPGA, Simulink (QUARC)

Design and simulation: SolidWorks, Fusion360, Anybody, OpenSim

Statistical analysis: SPSS

REFERENCES

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